

OAK TIRIAC 2 (STEJARII 2)



Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines energy optimisation and water-reduction measures applied across a 57,000 m² residential complex; sustainable materials, abundant daylight, biophilic design and infrastructure for separate waste collection; and electric-vehicle charging plus sport, relaxation and wellness services within the community. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Băneasa, northern Bucharest, Romania
Building use / function	Large premium residential complex with sport and wellness services
Project type (new build, refurbishment, extension)	New build
Plot area / Floor area	Approximately 57,000 m ²
Year / phase	Certified-project / phased development
Climate zone	Temperate continental
Thematic focus / innovation	Energy and water optimisation; sustainable materials; EV charging; biophilic design; daylight; separate waste collection
Investor / building owner	Țiriac Imobiliare
Architect	Not publicly disclosed in the consulted sources

Other involved stakeholders	Romania Green Building Council; sport, wellness, landscape and building-services teams
Contact person	Țiriac Imobiliare - https://www.tiriachimobiliare.ro/
Further information	https://www.rogbc.org/proiecte-certificate-ro gbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Oak Tiriac 2 (Stejarii 2) is a new build project in Băneasa, northern Bucharest, Romania. It functions as large premium residential complex with sport and wellness services. Its main best-practice contribution is Applies Green Homes criteria to a very large, amenity-rich residential campus in a wooded northern-Bucharest setting. The documented strategy brings together energy optimisation and water-reduction measures applied across a 57,000 m² residential complex, sustainable materials, abundant daylight, biophilic design and infrastructure for separate waste collection, and electric-vehicle charging plus sport, relaxation and wellness services within the community. The RoGBC toolkit gives the 57,000 m² area and qualitative measures; unit count and measured savings were not publicly identified. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Energy optimisation and water-reduction measures applied across a 57,000 m² residential complex.
2. Integrated systems and operation - Sustainable materials, abundant daylight, biophilic design and infrastructure for separate waste collection.
3. Wider value - Electric-vehicle charging plus sport, relaxation and wellness services within the community.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of [truth](#).